E-COMMERCE DATA ANALYSIS & INSIGHTS REPORT

Systematic Exploratory Data Analysis & Business Intelligence Summary

DATABASE NAME	TIMEFRAME COVERED	ANALYSIS TYPE	STATUS
ecommerce.db	Jan 2024 – Nov 2025	Full-Stack EDA	Complete & Validated

1. EXECUTIVE SUMMARY & KEY PERFORMANCE METRICS

This report synthesizes the core findings from the comprehensive Exploratory Data Analysis (EDA) conducted across all seven primary data entities: sales, products, user events, ratings, categories, order fulfillment, and geographic distribution.

ELECTRONICS REVENUE
$30.4M+
Top Grossing Category

FUNNEL CONVERSION
32.94%
View-to-Purchase Rate

CANCELLATION + RETURN
39.93%
Combined Friction Rate

TOP PRODUCT REVENUE
$858,086
Willow Hospital

2. PRODUCT CATEGORY PERFORMANCE BREAKDOWN

An evaluation of sales distribution across categories reveals heavy revenue concentration in high-ticket items, specifically **Electronics** and **Automotive**, while unit volume stays distributed in low-cost items like **Toys** and **Pet Supplies**.

CATEGORY	TOTAL PRODUCTS	UNITS SOLD	TOTAL REVENUE ($)	AVG RATING	REVENUE SHARE
Electronics	203	35,900	$30,439,067.72	3.56	39.2%
Automotive	193	36,002	$15,652,887.63	3.53	20.2%
Home & Kitchen	199	36,045	$6,900,475.43	3.54	8.9%
Sports	183	35,670	$5,956,274.42	3.53	7.7%
Clothing	213	40,299	$4,463,526.55	3.56	5.8%
Beauty	207	41,690	$3,622,282.98	3.57	4.7%
Toys	213	43,570	$2,608,012.50	3.55	3.4%
Pet Supplies	212	43,021	$2,284,032.69	3.59	2.9%
Books	194	38,713	$1,783,863.40	3.54	2.3%
Groceries	183	32,891	$540,396.51	3.54	0.7%

Category Insights

Electronics and Automotive jointly contribute nearly 60% of all gross sales value. Customer ratings remain uniform (3.53 - 3.59 stars) across all categories, proving that customer satisfaction challenges are systemic rather than category-specific.

3. CONVERSION FUNNEL & DROP-OFF DYNAMICS

Tracking user journey events (`view` → `cart` → `purchase`) provides clear metrics on engagement and cart abandonment points across the store ecosystem.

FUNNEL STAGE	UNIQUE USERS	CONVERSION RATE
1. Product Viewers	9,961	100.0% (Baseline)
2. Cart Additions	6,994	70.21% (View → Cart)
3. Completed Buyers	3,281	46.91% (Cart → Buy)

Funnel Diagnosis

While **70.21%** of viewers demonstrate intent by adding products to their carts, more than half (**53.09%**) abandon their carts prior to checkout completion. Addressing payment options and delivery fees can recapture significant lost revenue.

4. ORDER FULFILLMENT STATUS & OPERATIONAL FRICTION

Analyzing the status distribution across all historical orders highlights operational bottlenecks, with high cancellation and return proportions impacting net margins.

ORDER STATUS	TOTAL ORDERS	PERCENTAGE SHARE	TOTAL ORDER VALUE ($)	OPERATIONAL IMPACT
Shipped	4,113	20.57%	$24,238,665.56	In transit / Pending delivery
Returned	4,066	20.33%	$24,269,180.02	High Return Costs
Completed	4,021	20.11%	$24,197,125.58	Recognized Revenue
Cancelled	3,920	19.60%	$22,987,000.13	Lost Pipeline Value
Processing	3,880	19.40%	$23,494,716.66	Fulfillment queue

5. RATING DISTRIBUTION SUMMARY

Customer review sentiment reflects a strong core of satisfied customers alongside a notable block of neutral and dissatisfied buyers.

STAR RATING	TOTAL REVIEWS	PERCENTAGE
5 Stars	3,409	22.73%
4 Stars	4,624	30.83%
3 Stars	4,289	28.59%
2 Stars	2,084	13.89%
1 Star	594	3.96%

Quality Alert

Combined 1-Star and 2-Star reviews account for **17.85%** (2,678 reviews). Combined with 28.59% in 3-Star neutral ratings, over 46% of customers experience minor to major product dissatisfaction.

6. TOP 10 BEST-SELLING PRODUCTS BY REVENUE

The database exhibits key flagship products driving disproportionate gross revenue performance.

PRODUCT NAME	GROSS REVENUE ($)	MARKET POSITION
Willow Hospital	$858,086	#1 Revenue Leader
Pulse Security	$585,154	#2 Top Performer
Pulse Treat	$530,424	#3 Top Performer
GreenLeaf System	$529,486	#4 Key Revenue Driver
GreenLeaf Item	$446,018	#5 Key Revenue Driver
Willow Result	$425,403	#6 Top Performer
Orion Number	$417,621	#7 Top Performer
Harbor Cost	$416,758	#8 Top Performer
Willow Community	$406,280	#9 Top Performer
Orion Face	$405,789	#10 Top Performer

7. TOP GEOGRAPHIC MARKETS (BY CITY)

Evaluating geographic cluster performance by city identifies high-revenue regional centers for targeted marketing and logistics optimization.

CITY	TOTAL REVENUE	ORDERS	AOV ($)
North Michael	$21,598.73	23	$423.50
Lake Christopher	$13,483.00	18	$374.53
Port James	$12,668.18	14	$422.27
South Mary	$11,720.35	11	$355.16
East David	$11,551.78	11	$444.30

CITY	TOTAL REVENUE	ORDERS	AOV ($)
South Ashley	$11,207.84	11	$431.07
North Jennifer	$11,145.55	11	$301.23
New Nathaniel	$11,142.14	11	**$619.01**
East Beth	$11,136.58	11	$464.02
Johnsonchester	$10,633.38	10	$366.67

8. ACTIONABLE STRATEGIC RECOMMENDATIONS

1. Optimize Checkout & Reduce Cart Abandonment

Implement retargeting emails, clear shipping fees prior to the cart page, and 1-click checkout options to convert the 53% drop-off between cart additions and purchases.

2. Address High Return and Cancellation Rates (~40%)

Investigate log quality control and delivery SLA bottlenecks. Reducing cancellation/return rates from 40% down to industry standard (15-20%) will preserve millions in gross margin.

3. Expand Electronics & High-AOV Cities

Focus ad spend on high Average Order Value (AOV) cities like **New Nathaniel ($619 AOV)** and scale stock allocation for top revenue drivers in **Electronics** and **Automotive**.